

# AFL Data Dictionary & Target Definitions

## Dataset Summary

| Dataset                             | Unit of Analysis      | Purpose                                                  |
|-------------------------------------|-----------------------|----------------------------------------------------------|
| cleaned_round_by_round_stats_v2.csv | One player per match  | Player-level performance analysis and player prediction. |
| cleaned_team_matches.csv            | One team per match    | Match outcome prediction and team-level analysis.        |
| cleaned_seasonal_stats.csv          | One player per season | Historical player summaries and feature enrichment.      |

## Key Variables

| Variable       | Description                                          |
|----------------|------------------------------------------------------|
| player_id      | Unique player identifier.                            |
| team           | Team represented by the player or match.             |
| opponent       | Opposing team.                                       |
| match_date     | Match date.                                          |
| year           | AFL season.                                          |
| round          | Competition round.                                   |
| result         | Match outcome (Win/Loss/Draw).                       |
| goals          | Goals scored.                                        |
| kicks          | Total kicks.                                         |
| handballs      | Total handballs.                                     |
| disposals      | Kicks + handballs.                                   |
| fantasy_points | Overall fantasy performance score.                   |
| margin         | Point difference between the teams.                  |
| home_away      | Whether the team is playing at home (H) or away (A). |

## Prediction Targets

| Target                   | Problem Type          | Definition                                                                                                                    |
|--------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Match Winner             | Binary Classification | Predict whether a team wins or loses a match. Draws must be handled separately or excluded when constructing a binary target. |
| Top Goal Kicker          | Ranking               | Rank players within a match by goals scored; the highest scorer is the goal-kicking leader.                                   |
| Top Disposal Getter      | Ranking               | Rank players within a match by disposals; the player with the highest total is the disposal leader.                           |
| Fantasy Points Leader    | Ranking               | Rank players within a match by fantasy points; the highest-scoring player is the fantasy leader.                              |
| Match Margin (Secondary) | Regression            | Predict the point difference between the teams.                                                                               |

## Engineered Features

- Rolling averages over the previous 3 and 5 matches for goals, disposals and fantasy points.
- Career experience, representing the player's prior career-match experience available before the current match.
- Days since the player's previous match.
- Numeric competition round (round\_number).
- Team win streak entering the match.
- Historical head-to-head win rate entering the match.
- Home/away match status (home\_away).

**Leakage control: rolling and historical features use only information available before the current match. Rolling player features use shift(1), and team/history features are shifted so the current match is excluded.**

## Data Quality & Availability Notes

- Negative values in disposals, kicks and handballs were treated as invalid because these match statistics cannot be negative.
- Negative fantasy\_points were retained because fantasy scoring can legitimately produce negative scores.
- Venue familiarity was not included because venue information was not available in the player-level feature data used for the final feature set.
- Weather, player position and travel distance were unavailable and therefore excluded from the feature set.